1 Five Basic Maclaurin Series

Here is a summary of the Maclaurin series we worked out last time:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!} \qquad (\text{for } x \in \mathbb{R})$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \qquad (\text{for } x \in \mathbb{R})$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \qquad (\text{for } x \in \mathbb{R})$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} \qquad (\text{for } |x| < 1)$$

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \qquad (\text{for } |x| < 1)$$

2 New Maclaurin Series from Old

Use the list of basic Maclaurin series and their intervals of convergence to help you answer the following questions.

1.) Find the Maclaurin series for the function $f(x) = x\cos(x)$ using the Maclaurin series for $\cos(x)$ above. (*Hint*: Recall that if $\sum a_k$ is a convergent series and c a constant, then $\sum ca_k$ is a convergent series and $\sum ca_k = c\sum a_k$.)

2.) Use substitution to find the Maclaurin series for each of the following functions and find the interval of convergence.

(a) $\sin(3x)$

(b) e^{4x}

(c) $\cos\left(\frac{x}{2}\right)$

(d) $\dfrac{1}{e^{2x}}$

(e) $\ln(1+5x)$

(f) $\ln\left(1-\dfrac{x}{3}\right)$